

parameter to be consistent with the current observations. We also calculate the scattering cross section between the DM particle and nucleon and discuss the implication for the direct DM search experiments. We summarize our results in the section IV. Our notations and the formulas used in our analysis are listed in Appendix.

II. THE MINIMAL GAUGED $U(1)_{B-L}$ MODEL WITH Z_2 PARITY

The model is based on the gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y \times U(1)_{B-L}$. Additional fields besides the standard model fields are a gauge field Z'_μ of the $U(1)_{B-L}$, a SM singlet $B-L$ Higgs boson Ψ with two $U(1)_{B-L}$ charge, and three RH neutrinos N_i which are necessary for the gauge and gravitational anomaly cancellations. In describing the RH neutrinos, we use the four component representation of RH neutrino constructed from the Weyl spinor ν_{R_i} ,

$$N_i \equiv \begin{pmatrix} \nu_{R_i} \\ \epsilon \nu_{R_i}^* \end{pmatrix}, \quad (1)$$

For the two RH neutrinos, N_1 and N_2 , we assign Z_2 parity even, while odd for N_3 , so that the RH neutrino N_3 is stable and, hence, the DM candidate.

Due to the additional gauge symmetry $U(1)_{B-L}$, the covariant derivative for each fields is given by

$$D_\mu = D_\mu^{(SM)} - iq_{B-L} g_{B-L} Z'_\mu, \quad (2)$$

where $D_\mu^{(SM)}$ is the covariant derivative in the SM, and q_{B-L} is the charge of each fields under the $U(1)_{B-L}$ with its gauge coupling g_{B-L} .

Yukawa interactions relevant for the neutrino masses are given by

$$\mathcal{L}_{int} = \sum_{\alpha=1}^3 \sum_{i=1}^2 y_{\alpha i} \bar{L}_\alpha \tilde{\Phi} N_i - \frac{1}{2} \sum_{i=1}^3 \lambda_{R_i} \bar{N}_i \Psi P_R N_i + \text{h.c.}, \quad (3)$$

where $\tilde{\Phi} = -i\tau_2 \Phi^*$ for Φ being the SM Higgs doublet, and without loss of generality we have worked out in the basis where the second term in the right-hand-side is in flavor diagonal for RH neutrinos. Because of the Z_2 parity, the DM candidate N_3 has no Yukawa couplings with the left-handed lepton doublets.

The general Higgs potential for the $SU(2)_L$ doublet Φ and a singlet $B-L$ Higgs Ψ is generally given by

$$V(\Phi, \Psi) = m_1^2 |\Phi|^2 + m_2^2 |\Psi|^2 + \lambda_1 |\Phi|^4 + \lambda_2 |\Psi|^4 + \lambda_3 |\Phi|^2 |\Psi|^2. \quad (4)$$